

Synchra Prop Desk: Private, Portfolio-Margined Proprietary Trading on Arc

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Abstract

On-chain proprietary trading firms – smart contracts that fund a trader against pinned, non-retroactive risk rules and pay out on a passed evaluation – are a real, fast-growing category (HyperPNL and comparable platforms), built as a direct response to the CeFi prop-firm industry’s reputation for changing rules after a trader qualifies. Every example live today shares two structural limits: they run on transparent order books, so a funded trader’s strategy is visible to the firm and to competitors in real time, and they margin positions in isolation, wasting the capital efficiency a multi-market trader should get from hedged exposure. This paper shows that a prop desk is not a new protocol to build – it is a composition of primitives Synchra’s clearing kernel already defines: capability tokens for non-retroactive risk rules, the portfolio margin engine for cross-market capital efficiency, TEE-matched execution for strategy privacy, and the existing vault performance-fee path for payout. We formalize the model, compare it against the current on-chain prop-firm architecture, and give the lifecycle and risk analysis for both the firm and the trader.

Keywords: proprietary trading, funded trader programs, portfolio margin, capability tokens, private execution

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1 Introduction

A prop firm funds a trader with the firm’s capital, not the trader’s own, and keeps the majority of the profit in exchange for taking on the capital risk. The CeFi version of this industry has a well-known trust problem: a trader passes an evaluation against one set of rules, and the firm tightens them, disputes a payout, or changes terms after the fact, because the rules were never anything more than the firm’s word.

On-chain prop firms answer this directly. Rules – daily loss limit, maximum drawdown, profit target, minimum trading days – are written into a smart contract at account creation and cannot be changed while the account is open. HyperPNL and comparable platforms run this today: a two-phase evaluation (e.g., 10% profit target / 9% max drawdown / 5% daily loss limit in Phase 1, a lower target with the same risk bounds in Phase 2), with payout triggered by the contract and settled in seconds once a trader qualifies. This is a young category – roughly six months old at the time of writing – but it is real, live, and growing.

Every implementation of it today shares two limits that come from the venue it is built on, not from the prop-firm model itself:

- **Transparency.** These platforms run on public order books. A funded trader’s size, direction, and open positions are visible to anyone watching the chain – including competitors, and including the firm itself, which can see exactly how a trader is positioned before an evaluation resolves.
- **Isolated margin.** Positions are margined per-market. A funded trader running a hedged, multi-market book gets no capital-efficiency credit for the hedge – the same gross-margin waste Synchra’s core paper describes for retail traders applies identically here, just with the firm’s capital instead of the trader’s own.

This paper’s claim is narrow and specific: Synchra does not need a new subsystem to support prop trading. The clearing kernel’s existing primitives – capability tokens, portfolio margin, TEE-matched execution, and vault performance-fee settlement – already compose into a prop desk that keeps the non-retroactive-rules property the category is built around, while removing both structural limits above.

2 Background: The On-Chain Prop-Firm Pattern

2.1 Why rules are pinned, not just published

The defining feature of the category is not that risk rules exist – CeFi prop firms publish rules too – it is that the rules are enforced by code the firm cannot unilaterally change once a trader’s account is open. This is the entire trust proposition: a trader who passes evaluation under a specific drawdown limit knows that limit was fixed before they started, and no update, dispute process, or after-the-fact reinterpretation can move it.

2.2 Typical parameters

2.3 What the venue determines, not the model

Both limits identified in Section 1 are properties of the underlying exchange, not of the prop-firm contract layered on top of it. A prop-firm contract sitting on a transparent, isolated-margin perp DEX inherits that DEX’s transparency and margin inefficiency regardless of

Parameter	Typical range (2026 on-chain prop firms)
Profit target	5%–10% (often lower in a second phase)
Daily loss limit	4%–5%
Max drawdown	8%–10%, trailing
Minimum trading days	Firm-specific, sometimes waived
Payout settlement	On-chain, seconds, in USDC

Table 1: Representative risk parameters from current on-chain prop-firm platforms.

how well-designed its own risk-rule logic is. This is the gap Synchrona’s kernel closes – not by building a better prop-firm contract, but by being a venue where the prop-firm layer inherits privacy and portfolio margin for free.

3 The Prop Desk Model

A prop desk on Synchrona is a specific configuration of four primitives already defined in the core clearing kernel – nothing here is a new contract type.

Prop-firm requirement	Existing Synchrona primitive
Risk rules pinned at account creation, non-retroactive	Capability token – signed once by the firm’s owner over the scoped action fields, kernel-verified, cannot re-signed to tighten terms on an open account
Daily loss limit / max drawdown / position caps	Agent execution policy – max position size, max leverage, daily loss limit, cooldown periods, all kernel-enforced, not left to the trader’s own code
Capital efficiency across a hedged book	Portfolio margin engine – f_K prices a genuine hedge a margin credit (core paper, portfolio margin section) giving a funded trader real efficiency an isolated-margin venue cannot
Strategy privacy from the firm and competitors	TEE-matched execution – the firm sets and enforces risk rules without ever seeing the trader’s live position only settlement events are visible on-chain
Payout on passing evaluation / profit split	Vault performance-fee settlement path – structured the same mechanism as a strategy vault manager’s reused rather than rebuilt

Table 2: The prop desk is a composition, not a new subsystem.

Definition 3.1 (Prop account). *A prop account is a clearing account $A_{prop} = (\mathcal{B}, \mathcal{P}, \mathcal{H})$ funded entirely by the firm’s capital allocation, associated with a capability*

$$capability = \text{Sign}_{sk_{firm}}(H(trader_id, limits, expiry))$$

where limits encodes the evaluation’s risk parameters (Table 1) as kernel-enforced execution-policy fields, not as advisory terms. The signature is produced once, at account creation; the kernel accepts trader actions only while a valid, unexpired capability with these exact limits exists, and the firm has no on-chain action that modifies limits for an open account.

4 Lifecycle

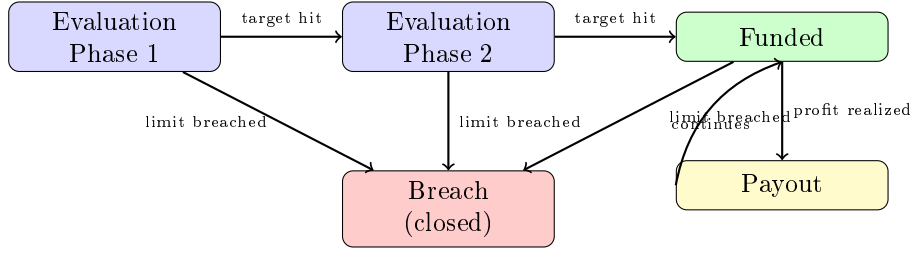


Figure 1: Prop account state machine. Every transition is a kernel-enforced check against the capability’s pinned limits (Section 3) – the firm observes state transitions, not the positions that caused them.

Evaluation phases. The trader operates the prop account under the pinned execution policy. Portfolio margin is computed exactly as for any other account (core paper, portfolio margin section) – a hedged evaluation trader gets the same f_K credit a funded trader would, which matters because evaluation performance under isolated margin materially understates what the same strategy could do once funded on a portfolio-margined venue.

Breach. A daily-loss or drawdown breach is a kernel-level check against C_{eff} and realized/unrealized PnL, evaluated the same way a liquidation threshold is (core paper, liquidation mechanism). The account closes; capital returns to the firm’s pool. No dispute process exists because no discretion exists – the check is a comparison against the pinned limits, not a judgment call.

Funded and payout. Passing both phases moves the account into funded status with the firm’s full allocation. Payout uses the existing vault performance-fee settlement path: realized profit above the firm’s retained share settles to the trader on the same schedule a vault would distribute returns, denominated in USDC, at Arc’s native settlement speed.

5 Privacy: What the Firm Sees

The central departure from the existing category is what the firm’s own contract is able to observe. On a transparent venue, the firm (and everyone else) can watch a trader’s live book while an evaluation is in progress – which means the firm can, in principle, see exactly how close a trader is to a limit before the trader does, and a competing firm can copy a strategy mid-evaluation. On Synchra, the firm’s capability token grants and enforces limits without granting visibility: the TEE matches the trader’s orders privately (core paper, private matching section), and the only things that reach the chain – and therefore the firm – are state transitions (Figure 1) and the collateral/PnL numbers needed to check a breach, not the position set that produced them. A trader is risk-limited by rules they cannot see removed, and private from a firm that cannot see their strategy. Neither property trades off against the other, because they are enforced by two different mechanisms – the capability token for limits, the TEE for confidentiality – that do not depend on each other.

6 Capital Efficiency

Consider a funded trader running a EURC/USDC hedge alongside a BTC/USDC directional position (once a second, correlated market exists on Synchra per the core paper’s roadmap).

Under isolated margin – the model every current on-chain prop firm uses – the trader posts full margin on both legs independently. Under Synchra’s portfolio margin, $f_K(\mathcal{P}) = \beta \sum \rho_{ij} s_i s_j$ prices the genuine offset as a margin credit. The firm’s capital backs the same trader activity with less locked collateral, which is a direct improvement in how much trading a fixed capital allocation can support – not a new feature, simply the core paper’s leverage moat applied to a funded account instead of a self-funded one.

7 Risk Analysis

Firm risk. The firm’s maximum loss on a given prop account is bounded by the pinned execution policy, enforced at the kernel level rather than trusted to the trader’s own discipline or code – the same enforcement model already described for agent accounts generally (core paper, agent security section). A compromised or malicious trader cannot exceed the daily loss limit or max drawdown any more than a compromised agent can exceed its capability’s scope elsewhere in the system.

Trader risk. The trader’s protection is that the limits cannot be tightened after the capability is signed – the core trust property the entire category exists to provide, here inherited directly from the capability-token design rather than re-implemented.

Shared risk. Both parties inherit the core paper’s stated risks generally: oracle dependency for breach/liquidation checks, TEE hardware trust for the privacy property, and smart contract risk in the kernel itself (core paper, Section 1.4 and security analysis). Nothing about the prop desk composition introduces a new risk category – it inherits exactly the kernel’s existing risk surface, which is the point of building it as a composition rather than new infrastructure.

8 Relation to the Core Protocol

This is an application-layer pattern, not a protocol change. It requires no new kernel contracts beyond what the core paper already specifies for agent accounts, portfolio margin, TEE matching, and strategy vaults. It depends on: agent accounts with capability tokens (core paper, agent-native section, MVP), the full portfolio margin engine (MVP), working TEE-matched execution (MVP), and institutional sub-account isolation for the firm’s capital pool (core paper roadmap, Phase 3). A minimal single-trader version – one firm account, one capability, no sub-account hierarchy – is reachable as soon as the MVP’s agent-account and portfolio-margin pieces exist; the multi-trader, multi-desk version follows institutional sub-accounts in Phase 3.

9 Conclusion

On-chain prop trading is a real category solving a real trust problem – non-retroactive, code-enforced risk rules – but every current implementation inherits its venue’s transparency and isolated-margin limits. Synchra does not need to build a competing prop-firm product; its clearing kernel already contains every primitive the category depends on, plus two properties – strategy privacy and portfolio margin – that no current on-chain prop firm has. A prop desk is simply what those primitives look like composed for this specific use case.